역함수 06

연습문제 풀이

연습문제 6.1

2. 다음 함수의 역함수를 구하여라.

(1) $f(x)=-4x^3-1$

$x=-4y^3-1$

$4y^3=-(x+1)$

$y^3=-\frac{1}{4}(x+1)$

$\therefore f^{-1}(x)=\sqrt[3]{-\frac{1}{4}(x+1)}$

(2) $f(x)=\frac{1-2x}{5x}$

$x=\frac{1-2y}{5y}$

$5xy=1-2y$

$(5x+2)y=1$

$\therefore f^{-1}(x)=\frac{1}{5x+2}$ $\left(\text{단, } x\neq-\frac{2}{5}\right)$

4. 다음 함수의 $(f^{-1})'(c)$를 구하여라.

(1) $f(x)=5x^5+4x^3,\ c=9$

$9=f(1)$ 이고, $f'(x)=25x^4+12x^2$이므로

정리6.4에 의하여

$(f^{-1})'(9)=\frac{1}{f'(1)}=\frac{1}{37}$

답: $\frac{1}{37}$

(2) $f(x)=x\ln x,\ c=2e^2$

$2e^2=f(e^2)$이고,

$f'(x)=1(\ln x)+x\frac{1}{x}=\ln x+1$이므로

정리6.4에 의하여

$(f^{-1})'(2e^2)=\frac{1}{f'(e^2)}=\frac{1}{2+1}=\frac{1}{3}$

답: $\frac{1}{3}$

연습문제 6.2

1. 다음 함수의 도함수를 구하여라.

(1) $y=e^{\sqrt{x+1}}$

$$y'=e^{\sqrt{x+1}}(\sqrt{x+1})'=(e^{\sqrt{(x+1}})(\frac{1}{2}(x+1)^{-\frac{1}{2}})$$

$$=\frac{1}{2\sqrt{x+1}}e^{\sqrt{x+1}}$$

(2) $e^{xy}+y=2$

$$e^{xy}\left(y+x\frac{dy}{dx}\right)+\frac{dy}{dx}=0$$

$$\frac{dy}{dx}(xe^{xy}+1)=-ye^{xy}$$

$$\therefore \frac{dy}{dx}=-\frac{ye^{xy}}{xe^{xy}+1}$$

(8) $y=e^{2x}\ln x$

$$y'=2e^{2x}\cdot\ln x+e^{2x}\cdot\frac{1}{x}$$

$$=e^{2x}\left(2\ln x+\frac{1}{x}\right)$$

2. 다음 적분을 구하여라.

(1) $\int\frac{e^{\sqrt{x}}}{\sqrt{x}}dx$

$$u=\sqrt{x}$$

$$\Rightarrow du=\frac{1}{2\sqrt{x}}dx$$

$$\Rightarrow \frac{1}{\sqrt{x}}dx=2du$$

$$\int\frac{e^{\sqrt{x}}d}{\sqrt{x}}dx=\int e^{u}\cdot 2du$$

$$=2\int e^{u}du=2e^{u}+c=2e^{\sqrt{x}}+c$$

(3) $\int\frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}dx$

$$u=e^{x}+e^{-x}$$

$$\Rightarrow du=(e^{x}-e^{-x})dx$$

$$\int\frac{e^{x}-e^{-x}}{e^{x}+e^{-x}}dx=\int\frac{1}{u}du$$

$$=\ln|u|+c=\ln|e^{x}+e^{-x}|+c$$

(7) $\int \frac{e^{-x}\ln(1+e^{-x})}{1+e^{-x}}dx$

$u=\ln(1+e^{-x})$

$\Rightarrow du=\frac{-e^{-x}}{1+e^{-x}}dx$

$\Rightarrow \frac{e^{-x}}{1+e^{-x}}dx=(-1)du$

$\int \frac{e^{-x}\ln(1+e^{-x})}{1+e^{-x}}dx=-\int u\,du$

$=-\frac{1}{2}u^2+c=-\frac{1}{2}\{\ln(1+e^{-x})\}^2+c$

(8) $\int \frac{e^{\tan x}}{\cos^2 x}dx$

$u=\tan x$

$\Rightarrow du=\sec^2 x dx=\frac{1}{\cos^2 x}dx$

$\int \frac{e^{\tan x}}{\cos^2 x}dx=\int e^{\tan x}\sec^2 x dx=\int e^u du$

$=e^u+c=e^{\tan x}+c$

연습문제 6.3

1. 다음 주어진 값을 구하여라.

(1) $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)=\frac{\pi}{3}$

(2) $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)=-\frac{\pi}{6}$

2. 다음 함수의 도함수를 구하여라

(1) $f(x)=\cos^{-1}(-3x)$

$$f'(x)=\frac{-1}{\sqrt{1-(-3x)^2}}\cdot(-3x)'=\frac{3}{\sqrt{1-9x^2}}$$

(3) $f(x)=\sin(\tan^{-1}x)$

$$f'(x)=\cos(\tan^{-1}x)\cdot(\tan^{-1}x)'=\frac{1}{1+x^2}\cdot\cos(\tan^{-1}x)$$

5. 다음을 적분하여라.

(1) $\int\frac{1}{9x^2+16}dx$

$$=\frac{1}{16}\int\frac{1}{\frac{9}{16}x^2+1}dx$$

$$=\frac{1}{16}\int\frac{1}{\left(\frac{3}{4}x\right)^2+1}d\left(\frac{3}{4}x\right)\left(\frac{4}{3}\right)$$

$$=\left(\frac{1}{16}\right)\left(\frac{4}{3}\right)\tan^{-1}\left(\frac{3x}{4}\right)+c$$

$$=\frac{1}{12}\tan^{-1}\left(\frac{3x}{4}\right)+c$$

(2) $\int \frac{1}{2x^2+4x+6}dx$

$$\frac{1}{2}\int \frac{1}{x^2+2x+3}dx = \frac{1}{2}\int \frac{1}{(x+1)^2+2}dx$$

$$= \frac{1}{4}\int \frac{1}{\frac{1}{2}\cdot(x+1)^2+1}dx = \frac{1}{4}\int \frac{1}{\left(\frac{x+1}{\sqrt{2}}\right)^2+1}dx$$

$$\left(u=\frac{x+1}{\sqrt{2}},\ du=\frac{1}{\sqrt{2}}dx,\ dx=\sqrt{2}du\right)$$

$$= \frac{1}{4}\int \frac{\sqrt{2}}{u^2+1}du$$

$$= \left(\frac{1}{4}\right)(\sqrt{2})\cdot \tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right)+c$$

$$= \left(\frac{\sqrt{2}}{4}\right)\cdot \tan^{-1}\left(\frac{x+1}{\sqrt{2}}\right)+c$$

(3) $\int \frac{1}{x\sqrt{2x^4-25}}dx$

$$= \int \frac{x}{x^2\sqrt{2x^4-25}}dx$$

$$= \int \frac{\frac{2\sqrt{2}}{5}x}{\frac{(2)\sqrt{2}}{5}x^2(5)\sqrt{\left(\frac{\sqrt{2}x^2}{5}\right)^2-1}}dx$$

$$\left(\because u=\frac{\sqrt{2}x^2}{5},\quad du=\frac{2\sqrt{2}}{5}xdx,\quad \int \frac{du}{u\sqrt{u^2-1}}=\sec^{-1}|u|+c\right)$$

$$= \frac{1}{2\times 5}\int \frac{du}{u\sqrt{u^2-1}}$$

$$=$$

$$= \left(\frac{1}{10}\right)\sec^{-1}|u|+c$$

$$= \left(\frac{1}{10}\right)\sec^{-1}\left|\frac{\sqrt{2}x^2}{5}\right|+c$$

(5) $\int \frac{e^{-x}}{1+e^{-2x}}dx$

$$((u=e^{-x},\ du=-e^{-x}dx)$$

$$-\int \frac{1}{1+u^2}du = -\tan^{-1}(u)+c$$

$$= -\tan^{-1}(e^{-x})+c$$

연습문제 6.4

1. 다음의 값을 구하여라.

(1) $\sinh(0)$

$$\sinh(0)=\frac{e^{0}-e^{-0}}{2}=0$$

(2) $\cosh(\ln 3)$

$$\cosh(\ln 3)=\frac{e^{\ln 3}+e^{-\ln 3}}{2}=\frac{3+\frac{1}{3}}{2}=\frac{5}{3}$$

(4) $\tanh(\ln x)$

$$\tanh(\ln x)=\frac{e^{\ln x}-e^{-\ln x}}{e^{\ln x}+e^{-\ln x}}=\frac{x-\frac{1}{x}}{x+\frac{1}{x}}=\frac{x^2-1}{x^2+1}$$

연습문제 6.5

1. $L'H\hat{o}pital$의 법칙을 이용하여 다음의 극한을 구하여라.

(1) $\lim_{x\to 0}\frac{\cos x-1}{x}$

$$\lim_{x\to 0}\frac{\cos x-1}{x}=\lim_{x\to 0}\frac{-\sin x}{1}=\frac{0}{1}=0$$

(3) $\lim_{x\to\infty}\frac{\tan\frac{1}{x}}{\frac{1}{x}}$

$$\lim_{x\to\infty}\frac{\tan\frac{1}{x}}{\frac{1}{x}}=\lim_{x\to\infty}\frac{\sec^2\frac{1}{x}\cdot(-x^{-2})}{(-x^{-2})}=\lim_{x\to\infty}\sec^2\frac{1}{x}=1$$

2. 제 2 $L'H\hat{o}pital$의 법칙을 이용하여 다음의 극한을 구하여라.

(2) $\lim_{x\to\infty}\frac{\ln(1+x)}{\ln x}$

$$\lim_{x\to\infty}\frac{\ln(1+x)}{\ln x}=\lim_{x\to\infty}\frac{\frac{1}{1+x}\cdot(1)}{\frac{1}{x}}=\lim_{x\to\infty}\frac{x}{1+x}=1$$